

1 What is resolution, with reference to microscopy?

- A The number of times the image of an object appears bigger.
 B The ability of a microscope to show internal parts of a specimen.
 C The shortest distance between separate objects that a microscope can show. ✓
 D The type of radiation used by a microscope to produce an image of a specimen.

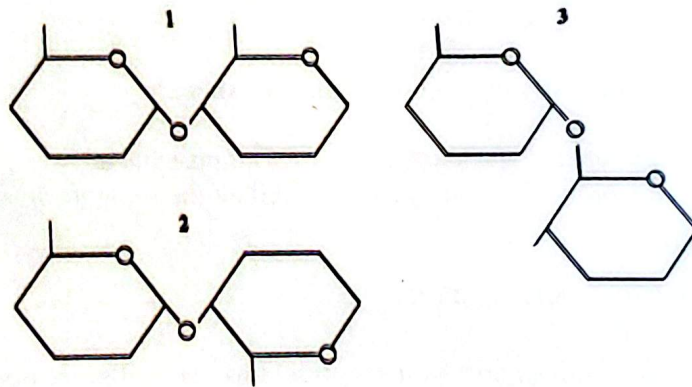
2 Which one is a function of the rough endoplasmic reticulum?

- A transports proteins — golgi
 B synthesis of steroids — smooth
 C makes lysosomes — ✓
 D internal processing

3 Which component of the cell surface membrane is responsible for cell-cell recognition?

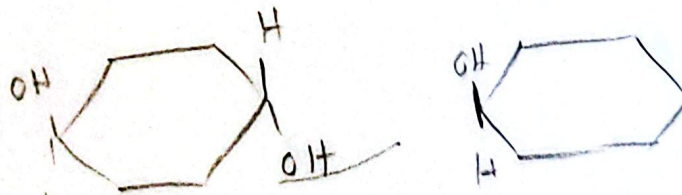
- A cholesterol
 B glycolipids ✓
 C phospholipids
 D enzymes

4 The diagrams show different types of bonds found in polysaccharides.

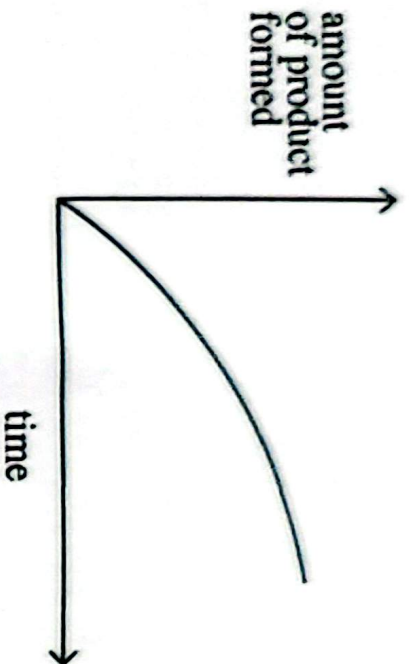


Which type(s) of bonds is/are found in amylose?

- A 1 only
 B 2 only
 C 1 and 3 ✓
 D 2 and 3 ✓



- 5 The graph shows the progress of starch digestion by the enzyme salivary amylase.



Why does the reaction slow down after some time?

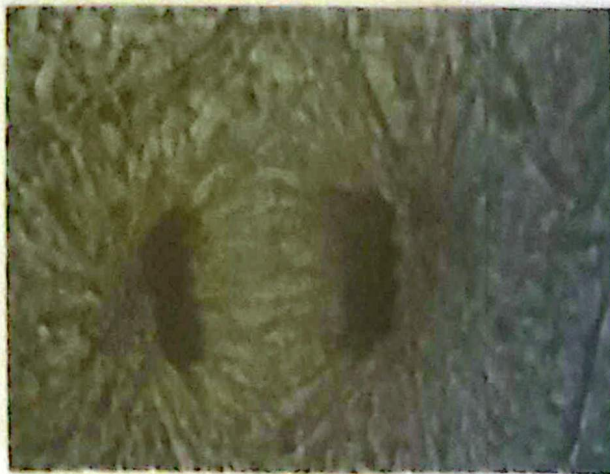
- A end product inhibition by maltose
- B salivary amylase is becoming denatured ~~✓~~
- C salivary amylase is becoming saturated with starch
- D the number of starch molecules is decreasing

- 6 Which row correctly describes a triglyceride?

(✓ = yes × = no)

	soluble in water	provides energy	produces water when oxidised
A	×	✓	✓
B	×	✓	×
C	✓	×	✓
D	✓	✓	×

- 7 The photomicrograph shows a stage in nuclear division.



Which stage is shown in the photomicrograph?

- A anaphase ✓
- B interphase ✗
- C metaphase
- D telophase ✗



- 8 A parent cell contains a nucleus with diploid number of 8.
How many possible types of gametes can be produced from it as a result of independent assortment?

- A 4
- B 8
- C 16 ✓
- D 32

$$2n = 8$$

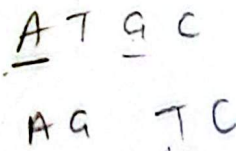
$$n = 4$$

$$2^4$$

An analysis of DNA from cells in a human body showed that thymine makes up 15% of the nucleotide bases.

What would be the percentage composition of guanine in the DNA?

- A 15
- B 35 ✓
- C 70
- D 85



- 10 Semi conservative replication of DNA means

- A one strand is copied continuously and the other staggered during replication.
- B the new molecule has alternating new and old bands.
- C there is a new and old strand in the molecule after replication. ✓✓
- D half of the molecules are old and half are new after replication.



11 Which one is an earliest stage during protein synthesis?

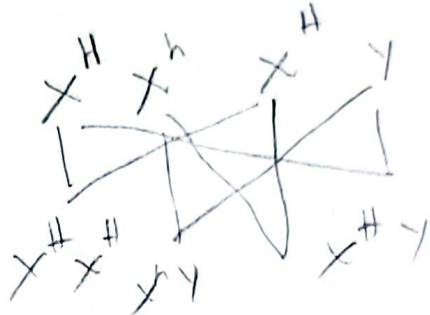
- A translation of the triplet code on mRNA to amino acids
- B attachment of the ribosome to mRNA
- C amino acid activation and attachment to tRNA
- D conversion of the DNA triplet code to mRNA ✓✓

12 Which enzyme is **not** needed in the production of recombinant DNA from mRNA?

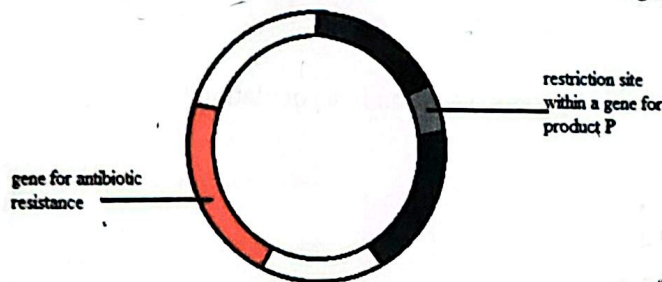
- A DNA ligase
- B reverse transcriptase
- C restriction endonuclease
- D RNA polymerase ✓✓

~~X~~ 13 Which one is **not** a benefit of genetic fingerprinting?

- A Reducing incidence of violent crime.
- B Solving parental disputes and cases of incest.
- C Reducing inbreeding of endangered species.
- D Resolving disputes over family relationships.



~~X~~ 14 The diagram shows a plasmid that can be used as a vector for transforming bacterial cells in the production of a biological product. The plasmid has two marker genes.



Which statement correctly describes the use of the marker genes in identifying transformed cells.

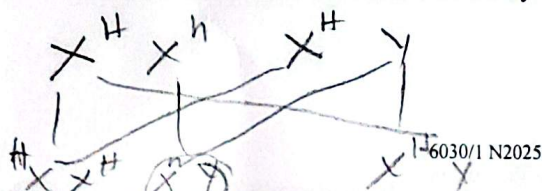
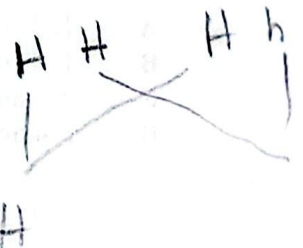
- A Transformed cells are resistant to the antibiotic.
- B Transformed cells produce product P.
- C Transformed cells are not resistant to the antibiotic.
- D Transformed cells will not produce product P.

~ (14)

H H

~~X~~ 15 A child is diagnosed with a rare genetic disease. Neither parent has the disease. How might the child have inherited the disorder?

- A The disorder is due to an environmental factor.
- B The disorder is dominant and was carried by a parent.
- C The disorder is sex linked and inherited only from the father. ✓
- D The disorder is recessive and carried by both parents ✓



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[Turn over

16 A trait controlled by four alleles is said to have

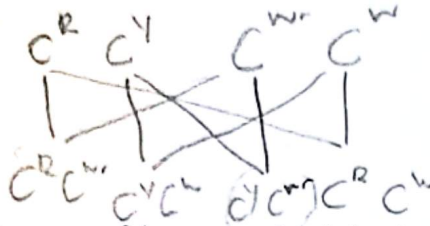
- A incomplete dominance.
- B complete dominance.
- C homologous alleles.
- D multiple alleles. ✓

17

In garden peas, *Pisium sativum*, round seed is dominant over wrinkled seed and yellow cotyledon is dominant over white cotyledon.

In a cross between heterozygous plants for both traits, what is the probability of producing wrinkled seeds with yellow cotyledons?

- A 0.0625
- B 0.1875
- C 0.5625
- D 0.7500



18 The following are some of the causes of variation in a population.

1. crossing over during meiosis ✓
2. different environmental conditions
3. random mating
4. mutation

Which of these lead to genetic variation in a population?

- A 1, 2 and 4
- B 1, 2 and 3
- C 1, 3 and 4 ✓
- D 2, 3 and 4

19 Some events that may occur within a population and its environment are:

1. gene mutation ✓
2. selection pressure ✓
3. predation of prey
4. environmental shift ✓
5. change of seasons

Which events are involved in the formation of a new species?

- A 1, 2 and 3
- B 1, 2 and 4 ✓
- C 2, 3 and 5
- D 3, 4 and 5



20 Antimycin is an insecticide which inhibits the electron transport chain.

Which row correctly shows the effect of antimycin?

	lactic acid	oxygen consumption	Krebs cycle
A	increases	increases	decreases
B	decreases	decreases	increases
C	increases	decreases	decreases
D	decreases	increases	decreases

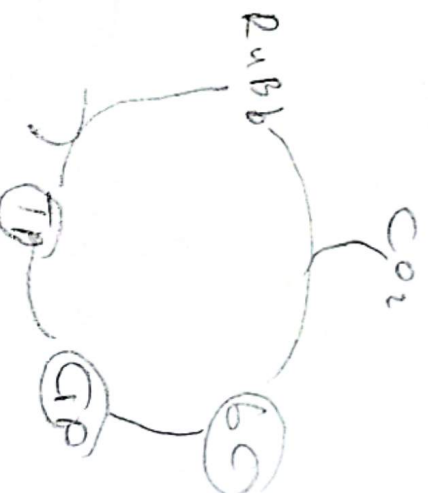
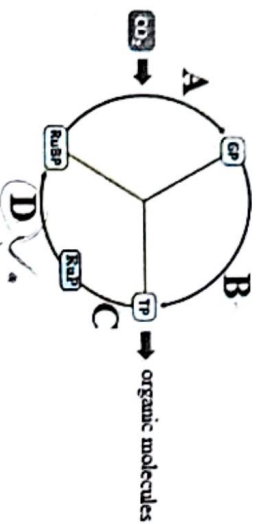
1.12 → 0.7
1.10 → 0.7
0.9 → 0.7
0.17

21 What causes the respiratory quotient (RQ) of germinating seeds to decrease from 1.0 to 0.7 with time?

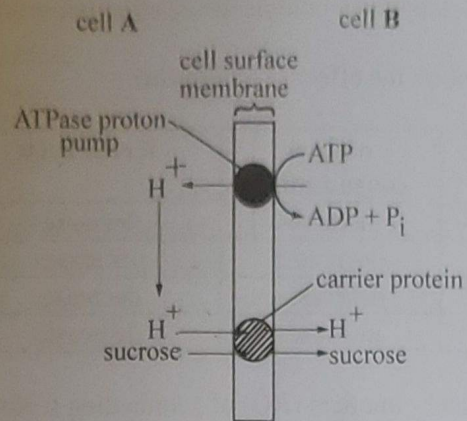
- A the respiratory substrate changes from carbohydrate to lipid
- B the respiratory substrate changes from lipid to carbohydrate
- C aerobic respiration replaces anaerobic respiration
- D anaerobic respiration replaces aerobic respiration

22 The diagram shows the Calvin Cycle.

Which step, A, B, C or D, uses both ATP and NADPH?



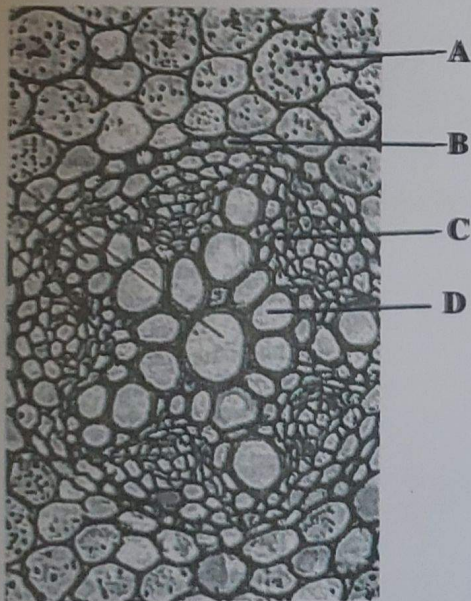
- 23 The diagram represents the loading of sucrose.



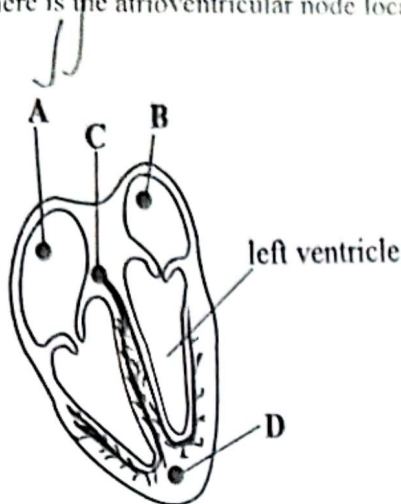
What are the identities of cell A and cell B?

	cell A	cell B
A	mesophyll	sieve tube element ✓
B	mesophyll	companion
C	companion	sieve tube element
D	companion	mesophyll

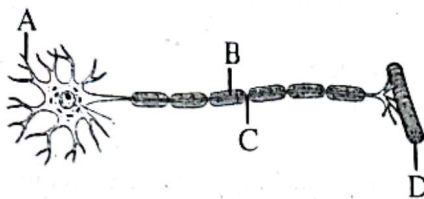
- 24 The micrograph shows a section through a root.
Which labelled tissue contains the Casparian strip?



- 25 The diagram below shows the section through the human heart. Where is the atrioventricular node located?



- 26 Which statement best defines systolic blood pressure?
- A the maximum blood pressure in the right ventricle
 - B the maximum blood pressure in the arteries
 - C the blood pressure in the arteries
 - D the blood pressure in the left ventricle at the end of contraction
- 27 Which of the following statements, about hyper-polarisation, is correct?
- A It is caused by the generation of an action potential.
 - B It is a result of a delay in the closing of sodium gates.
 - C It is because of a delay in the closing of potassium gates. ✓
 - D It is caused by the depolarisation of the excited membrane.
- 28 The diagram shows a motor neurone.
- Which part, A, B, C or D, allows the formation of a local circuit?



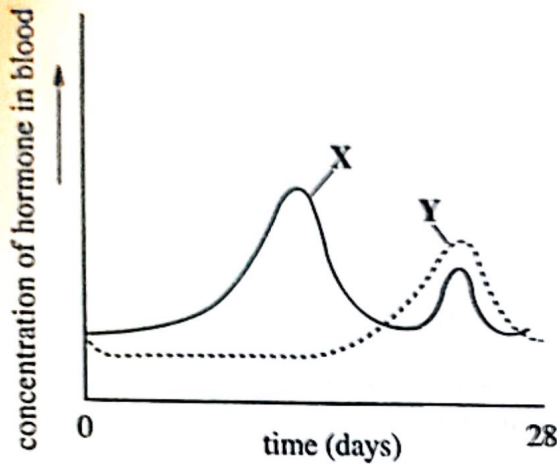
- 29 Which enzyme, secreted by a sperm, helps in its penetration into the ovum?
- A amylase
 - B catalase
 - C invertase
 - D protease. ✓



30 Which hormone is **not** secreted by the placenta during pregnancy?

- A chorionic gonadotrophin ✓
- B luteinising hormone
- C oestrogen
- D progesterone

31 The graph shows changes in the concentration of hormones X and Y in the blood during the menstrual cycle.



Which of the following correctly identifies hormones X and Y?

- | | hormone X | hormone Y |
|---|--------------|--------------|
| A | LH | oestrogen ✓ |
| B | oestrogen | FSH |
| C | oestrogen | progesterone |
| D | progesterone | oestrogen |

32 Release of untreated sewage into rivers decreases

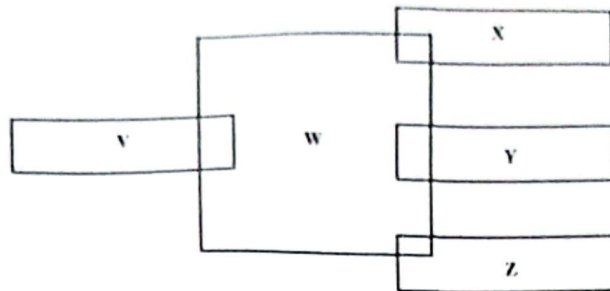
- A carbon dioxide concentration.
- B nitrate concentration.
- C phosphate concentration.
- D oxygen concentration.

33 Which agency or organisation in Zimbabwe directly involves the community in ecological conservation efforts?

- A CAMPFIRE ✓✓
- B UNESCO
- C EMA
- D WWF



- 34 Which one is a diagnostic feature of organisms in the phylum Cnidaria?
- A they are triploblastic
 - B possess radial symmetry ✓
 - C exist as polyps or medusae
 - D exhibit polymorphism
- 35 The diagram shows one way of classifying organisms into the five kingdoms.



Which kingdom is represented by the box labelled W?

- A fungi
 - B plantae ✓
 - C prokaryotae
 - D protista
- 36 Which taxon has the **lowest** diversity of living organisms?
- A genus ✓✓
 - B phylum
 - C order
 - D family
- 37 Which condition explains the high prevalence of malaria in the tropical regions?
- A high temperature ✓✓
 - B high wind speed
 - C low rainfall
 - D low humidity



- 38 The table shows information on diseases and factors affecting distribution. Which row is correctly matched?

	disease	factor affecting distribution
A	coronary heart disease	consumption of unsaturated fats
B	Ebola	human interaction with primates ✓
C	HIV/AIDS	poor sanitary conditions ✗
D	malaria	overcrowding ✗

- 39 Which disease is linked to a sedentary lifestyle?
- A cholera
 - B Ebola
 - C sickle cell anaemia
 - D coronary heart disease
- 40 Which disease has the same global distribution as malaria?
- A cholera
 - B coronary heart disease
 - C sickle cell anaemia ✓
 - D ebola